

Econometrics I

Classical Normal Regression Model: Hypothesis Testing

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Introduction

- I proceed with the construction of hypothesis tests about the population parameter β .
- I begin with a formal description of what hypothesis testing is.

Hypothesis Testing: definitions

Definition 1

A **Hypothesis** is a statement about a population parameter.

This definition is rather general, but the important point is that a hypothesis makes a statement about the population. The goal of the hypothesis test is to decide, based on a sample from the population, which of **two complementary hypothesis is true**.

Definition 2

The two complementary hypotheses in a hypothesis testing problem are called the **null hypothesis** and the **alternative hypothesis**. They are denoted by H_0 and H_1 , respectively.

Notation

If θ denotes a population parameter, the general format of the null and alternative hypothesis is $H_0 : \theta \in \Theta_0$ and $H_1 : \theta \in \Theta_0^c$, where Θ_0 is a subset of the parameter values that θ can take and Θ_0^c is the complement set of Θ_0 .

Example: Testing the efficacy of a drug.

Suppose that θ denotes the average change in a patient's blood pressure after taking a drug. A researcher might be interested in testing $H_0 : \theta = 0$ versus $H_1 : \theta \neq 0$. In this example, θ can be assumed to take any possible real number (the drug be effective, not effective, or even damaging); hence, $\Theta_0 = \{0\}$ and $\Theta_0^c = \mathbb{R} \setminus \{0\}$.

The null hypothesis states that, on average, the drug has no effect on blood pressure, and the alternative hypothesis states that there is some effect.

Example: Proportion of defective items

A consumer might be interested in the proportion of defective items produced by a supplier. If θ denotes the proportion of defective items, the consumer might wish to test $H_0 : \theta \geq \theta_0$ versus $H_1 : \theta < \theta_0$. In here, the value θ_0 is the maximum acceptable proportion of defective items. Since θ is a proportion, it is assumed to take values on the interval $[0, \infty)$, thus $\Theta_0 = [\theta_0, \infty)$ and $\Theta_0^c = [0, \theta_0)$. In here, the null hypothesis states that the proportion of defectives is unacceptably high.

How do we construct an hypothesis test?

In a hypothesis testing problem, after observing the sample, the researcher must decide either to accept H_0 as true or to reject H_0 as false and decide that H_1 is true.

Definition 3

A **hypothesis testing procedure** or **hypothesis test** is a rule that specifies

- 1 For which sample values the decision is made to accept H_0 as true.
- 2 For which sample values H_0 is rejected and H_1 is accepted as true.

The subset of the sample space for which H_0 will be rejected is called the **rejection region** or **critical region**. The complement of the rejection region is called the **acceptance region**.

Typically, a hypothesis test is specified in terms of a **test statistic** W , which is a function of the sample. You have seen examples of test statistics: the sample mean, the sample variance, and of course, the OLS estimators $\hat{\beta}$ and $\hat{u}'\hat{u}$ (the sum of squared residuals).

Remark

Sometimes the distinction is made between “rejecting H_0 ” and “accepting H_1 ”. In the first case, there is nothing implied about what state the researcher is accepting, only that the case defined by H_0 is rejected. Similarly, a distinction can be made between “accepting H_0 ” and “not rejecting H_0 ”. The first phrase implies that the researcher is willing to accept as true the case specified by H_0 , while the second phrase implies that the researcher does not believe H_0 but does not have the evidence to reject it. Be mindful of these distinctions.

Methods of Evaluating Tests

When deciding to accept or reject a null hypothesis H_0 , a researcher might be making a mistake. Usually, hypothesis tests are evaluated and compared through their probabilities of making mistakes.

A hypothesis test of $H_0 : \theta \in \Theta_0$ versus $H_1 : \theta \in \Theta_0^c$ might make one of two types of errors.

- 1 If $\theta \in \Theta_0$ but the hypothesis test incorrectly decides to reject H_0 then this is called a **Type I error**.
- 2 If, on the other hand, $\theta \in \Theta_0^c$ but the test decides to accept H_0 , a **Type II error** has been made.

		Decision	
		Accept H_0	Reject H_0
Truth	H_0	Correct Decision	Type I error
	H_1	Type II error	Correct Decision

Methods of Evaluating Tests

- Suppose we have a test statistic W , and for any “event” A , let $\Pr_{\theta}(W \in A)$ denote the probability that the values of W fall in A as a function of the value of θ . this probability is calculated using the **sampling distribution** of W .
- Let R denote the rejection region for a hypothesis test, i.e., R will specify for what observed values of W the test rejects H_0 .
 - 1 The probability of Type I error is:
 $\Pr_{\theta}(W \in R | \theta \in \Theta_0)$ = The probability that $W \in R$ given that H_0 is true.
 - 2 The probability of a Type II error is:
 $\Pr_{\theta}(W \in R^c | \theta \in \Theta_0^c)$ = The probability that $W \notin R$ given that H_1 is true.
- If the switch from R to R^c in (2) above confuses you, note that $\Pr_{\theta}(W \in R^c | \theta \in \Theta_0^c) = 1 - \Pr_{\theta}(W \in R | \theta \in \Theta_0^c)$.
- The function $\Pr_{\theta}(W \in R)$ defined for all possible values of θ is called the **power function of the test**.

$$\Pr_{\theta}(W \in R) = \begin{cases} \text{Probability of a Type I error} & \text{if } \theta \in \Theta_0 \\ 1 - \text{Probability of a Type II error} & \text{if } \theta \in \Theta_0^c \end{cases} \quad (1)$$

The power function

Definition 4

The **power function** of a hypothesis test with rejection region R is the function of θ defined by $\Pi(\theta) = \Pr_{\theta}(W \in R)$ where $\Pr_{\theta}(W \in R)$ is given by equation (1).

- The ideal power function would be 0 for all $\theta \in \Theta_0$ and 1 for all $\theta \in \Theta_0^c$. In general, this ideal cannot be attained. In fact, for a fixed sample size, it is usually impossible to make both types of error probabilities arbitrarily small.
- Explicit calculation or approximation of a power function is an advanced topic, which is for a more advanced course.
- In searching for a good test, it is common to restrict consideration to tests that control the Type I probability at a specified level, and within this class of tests we then choose tests that have Type II error probability as small as possible.

The size of the test

- In keeping with the ideas above, we will now focus on fixing the Type I probability at a *specified level*.
- We will also focus on null hypothesis for which Θ_0 is a singleton, i.e. $\Theta_0 = \{\theta_0\}$ so that the null hypothesis can be written as $H_0 : \theta = \theta_0$. We now have the following definition.

Definition 5

Consider null hypothesis of the form $H_0 : \theta = \theta_0$. Then for $0 \leq \alpha \leq 1$, a test with power function $\Pi(\theta)$ is a **size α test** if

$\Pi(\theta_0) = \Pr_{\theta}(W \in R | \theta = \theta_0) = \text{Probability of rejecting } H_0, \text{ i.e. } W \in R, \text{ when } H_0 \text{ is true} = \alpha.$

The size of the test is also called its **significance level**.

- Typical choices for the size of the test are $\alpha = 0.01, 0.05$ and 0.10 .

Practical considerations

- By fixing the size of the test, the researcher is controlling only for the Type I error probabilities, not the Type II error probabilities.
- When using this approach, you should specify the null and alternative hypothesis so that it is most important to control the Type I error probability.
- For instance, if you wish for your results to give support to a particular hypothesis, but you do not want to make the assertion unless the data really do convincing support, you can set the test up so that the alternative hypothesis is the one that you expect the data to support, and hope to prove it.
- The alternative hypothesis is sometimes called the research hypothesis in this context.
- By using a size α test with a small α , you are guarding against saying the data support the research hypothesis when it is false.

Hypothesis Testing in the CNR model.

- Recall that under assumptions MLR1–MLR6 we have that for the model $\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \mathbf{u}$, where $\mathbf{X}$ includes a column of ones, i.e., the model has a constant, we have that

$$\mathbf{y}|\mathbf{X} \sim N(\mathbf{X}\boldsymbol{\beta}, \sigma^2\mathbf{I}), \quad \text{and} \quad \text{rank}(\mathbf{X}) = k + 1$$

- In this set of slides, we will assume that σ^2 is known.

Test on a single parameter

- Suppose we have a hypothesis about the j th regression coefficient. Specifically:

- 1 Suppose the null hypothesis is $H_0 : \beta_j = \beta_j^\circ$ and,
- 2 suppose that the alternative is either one of the following options (only one)
 - a $H_1 : \beta_j \neq \beta_j^\circ$, this is called a **two-sided alternative**.
 - b $H_1 : \beta_j < \beta_j^\circ$, this is called a **one-sided left alternative**.
 - c $H_1 : \beta_j > \beta_j^\circ$, this is called a **one-sided right alternative**.

- The test statistic will be:

$$z_j^\circ = \frac{\hat{\beta}_j - \beta_j^\circ}{\sigma_{\hat{\beta}_j}}, \quad (2)$$

where $\hat{\beta}_j$ is the OLS estimate of β_j , and $\sigma_{\hat{\beta}_j} = \sigma \sqrt{q^{jj}}$ is the standard deviation of $\hat{\beta}_j$

- Recall property D3A in the slides CNR-model, page 9: $z_j = \frac{\hat{\beta}_j - \beta_j}{\sigma_{\hat{\beta}_j}} \sim N(0, 1)$.
- Then, **under the null hypothesis being true**, the test statistic z_j° in (2) has a standard normal distribution.

- To construct the rejection region, we focus on the specific type of alternative hypothesis under consideration and the size of the test we want to have.
- Remember, the size of a test is the *probability of a type I error*.
- Based on our previous discussion, we will fix the size of the test at the level α , and using this, we will construct the rejection region.
- In general, the rejection region R will satisfy $\Pr(z_j^0 \in R | \beta_j = \beta_j^0) = \alpha$. The probability calculation here is obtained using the standard normal distribution, since under the null hypothesis being true, z_j^0 is standard normal. Notice also that we are calculating the probability of observing a test statistic falling in the rejection region, and thus rejecting the null, when the null is in fact true and we are equating this probability to the size of the test (prob. of making a type I error) which we fixed beforehand.

- After fixing α , it is still not clear how to construct the rejection region. At this point is where we turn to the power function for guidance. This points towards the following rejection regions:
 - a** For $H_1 : \beta_j \neq \beta_j^\circ$, the type I error is:
 $\alpha = \Pr(|z_j|^\circ > z_{\alpha/2}) = \Pr(z_j^\circ > z_{\alpha/2} \text{ OR } z_j^\circ < -z_{\alpha/2})$, where $z_{\alpha/2}$ is the critical value of a standard normal distribution which gives an area under the tail of the distribution equal to $\alpha/2$.¹
 - b** For $H_1 : \beta_j < \beta_j^\circ$, the type I error is: $\alpha = \Pr(z_j^\circ < -z_\alpha) = \alpha$ where $-z_\alpha$ is the critical value of the standard normal distribution which gives an area under the *left tail of the distribution* equal to α .
 - c** For $H_1 : \beta_j > \beta_j^\circ$, the type I error is: $\alpha = \Pr(z_j^\circ > z_\alpha) = \alpha$ where z_α is the critical value of the standard normal distribution which gives an area under the *right tail of the distribution* equal to α .
- Notice that for two-sided tests, we divide the size α by 2 to get the critical value.

¹Notice how a two-sided test rejection region with level α relates to a confidence interval with confidence level $1 - \alpha$.

Discussion

- Consider a two sided alternative hypothesis and suppose that the size of the test is 5%. Then the critical value is $z_{0.025} = 1.96$.
- suppose a sample has $|z_j^\circ| > 1.96$. If the null is true, then a low-probability event has occurred. The probability of the event is so low that its occurrence is taken to be evidence against the null; so the decision is to reject the null. The point estimate of $\hat{\beta}_j$ is so far from the hypothesized parameter value β_j° that it is implausible that $\hat{\beta}_j$ was drawn from a distribution with expected value β_j° .
- However, finding a sample with $|z_j^\circ| \leq 1.96$ is not surprising when the null is true, so then the decision is to accept the null.
- Some remarks:
 - ▶ Rejection of the null is not proof the null is false. After all, in the calculations it is explicit that we are constructing the rejection region so as to fix the probability of the type I error, i.e., the size α of the test is fixed, and loosely speaking, when the null is true, in $\alpha\%$ of the samples the decision will be to “reject the null.”
 - ▶ Acceptance of the null is not proof that the null is true. Other values of β_j° would have been acceptable. In fact, for two-sided alternatives, any number β_j° that lies in the $1 - \alpha$ confidence interval and is used as a value in the null hypothesis, would also have led to accept it.

Testing hypothesis about single linear combination of β

- A procedure like the previous one also works for **single linear combinations of β** .
- Let $\theta = \mathbf{h}'\beta$, and $t = \mathbf{h}'\hat{\beta}$, where $\mathbf{h}$ is a $(k+1) \times 1$ nonrandom vector.
- We can rewrite property D2 in the slides CNR-model.pdf as $z = (t - \theta)/\sigma_t \sim N(0, 1)$, where $\sigma_t^2 = \sigma^2 \mathbf{h}'\mathbf{Q}^{-1}\mathbf{h}$.
- Thus, if the null is $H_0 : \theta = \theta^\circ$, then under the null being true, $z = (t - \theta^\circ)/\sigma_t \sim N(0, 1)$ and the procedure detailed in the previous slides remains the same.

Test on a set of parameters

- Now suppose we want to have a **joint null hypothesis** about β , specifically about p linear functions of β .
- Let $\theta = \mathbf{H}\beta$, where the $p \times (k + 1)$ nonrandom matrix $\mathbf{H}$ has rank p .
- The null hypothesis now is $H_0 : \theta = \theta^\circ$, where θ° is a specific numerical $p \times 1$ vector.
- We will only focus on two-sided hypothesis tests.
- Let $\mathbf{t} = \mathbf{H}\hat{\beta}$.
- The test statistic will be

$$w^\circ = \frac{(\mathbf{t} - \theta^\circ)' \mathbf{D}(\mathbf{t} - \theta^\circ)}{\sigma^2} \quad (3)$$

- Now, recall property D4 in CNR-model1.pdf, page 8: $w = (\mathbf{t} - \theta)' \mathbf{D}(\mathbf{t} - \theta) / \sigma^2 \sim \chi^2(p)$.
- Then, **under the null hypothesis being true**: the test statistic $w^\circ \sim \chi^2(p)$.

Constructing the rejection region for a test on a set of parameters

- Since w° is a quadratic form, it is always positive.
- Thus, for a fixed-size α the calculation of the critical region will use the tail probabilities of a chi-squared dist. with p degrees of freedom:

$$\alpha = \Pr(w^\circ > c_{p,\alpha})$$

where $c_{p,\alpha}$ is the critical value of a $\chi^2(p)$ which gives an area under the right-tail of the distribution equal to α .

- Intuitively, if a sample has $w^\circ > c_{p,\alpha}$, then if the null is true, then a low-probability event has occurred. The probability of this event is so low that its occurrence is taken to be evidence against the null.
- Heuristically, the point estimate $\mathbf{t}$ is so far from the hypothesized value θ° that it is deemed implausible that $\mathbf{t}$ has in fact been drawn from a distribution with expectation θ° .
- But finding a sample with $w^\circ \leq c_{p,\alpha}$ is not surprising when the null is true.
- The same remarks apply: in $\alpha\%$ of the cases, a test of this form will still be making the wrong decision, i.e., rejecting the null when it is true.